

Patient Management Portal - Integration Architecture & API C

Confidential - Internal Use Only

Version 1.0

1. System Context

The HCP Patient Portal connects patients to healthcare services, integrating with multiple clinical and administrative systems.

External Systems: Epic/Cerner EHR (patient records via HL7 FHIR R4), Surescripts (e-prescribing), Zoom Healthcare (telehealth), Stripe (patient payments), Onfido (identity verification), Synthea (test data generation).

Internal Services: Patient Service (profiles, auth), Scheduling Service (appointments, availability), Clinical Service (records, labs, prescriptions), Billing Service (claims, payments), Notification Service (email, SMS, push), Telehealth Service (video sessions).

2. HL7 FHIR R4 Integration

REQ-HCP-060: Patient demographics synchronized with EHR via FHIR Patient resource.

Bi-directional sync every 15 minutes.

REQ-HCP-061: Appointment resources created in EHR when patient books via portal. FHIR Appointment resource with status lifecycle: proposed -> booked -> arrived -> fulfilled -> cancelled.

REQ-HCP-062: Lab results consumed from EHR via FHIR DiagnosticReport and Observation resources. LOINC codes used for standardized test identification.

REQ-HCP-063: Medication lists synchronized via FHIR MedicationRequest resource. Includes RxNorm codes for drug identification.

REQ-HCP-064: SMART on FHIR authorization for third-party app access to patient data with granular scope control.

3. Surescripts E-Prescribing Integration

REQ-HCP-065: NewRx, RxRenewal, and CancelRx message types supported via Surescripts NCPDP SCRIPT 2017071 standard.

REQ-HCP-066: Medication history query (RxHistoryRequest) retrieves 12-month fill history from all pharmacies.

REQ-HCP-067: Real-time formulary and benefit check (RxBenefitCheck) shows patient cost estimate before prescribing.

REQ-HCP-068: EPCS (Electronic Prescribing of Controlled Substances) supported with DEA-compliant two-factor authentication.

4. Insurance Payer EDI Integration

REQ-HCP-069: Eligibility verification via X12 EDI 270/271 transaction. Real-time and batch modes supported.

REQ-HCP-070: Claim submission via X12 EDI 837 Professional (837P). Supports primary and secondary payer billing.

REQ-HCP-071: Remittance advice (ERA) received via X12 EDI 835. Auto-posted to patient accounts.

REQ-HCP-072: Claim status inquiry via X12 EDI 276/277 for tracking adjudication progress.

5. Sequence Flow: Telehealth Appointment

1. Patient Portal -> Scheduling Service: POST /api/appointments {provider_id, date, time, type: 'telehealth'}

2. Scheduling Service -> EHR (FHIR): POST /Appointment {status: 'booked', serviceType:

'telehealth'}

3. Scheduling Service -> Notification Service: Send confirmation email and calendar invite
4. Notification Service -> Patient: Email with 'Join Video Visit' link + SMS reminder at T-2h
5. Patient Portal -> Telehealth Service: POST /api/sessions/join {appointment_id, patient_id}
6. Telehealth Service -> Zoom Healthcare API: POST /meetings {topic, start_time, settings: {waiting_room: true}}
7. Zoom API -> Telehealth Service: {meeting_id, join_url, host_url}
8. Telehealth Service -> Patient Portal: {join_url} -> Patient enters virtual waiting room
9. Provider EHR -> Telehealth Service: Provider joins as host via host_url
10. Telehealth Service: Video session active, recording with consent stored to Azure Blob encrypted
11. Provider -> Clinical Service: POST /api/clinical-notes {appointment_id, soap_note, orders}
12. Clinical Service -> EHR (FHIR): POST /Encounter, POST /Observation, POST /MedicationRequest